

Jocelyn Chalendar

Embedded Software Engineer

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Professional Summary

Embedded Software developer with 10 years of experience, enabling product release from PoC to mass production, ensuring hardware validation and industrialization.

With a strong experience on critical applicative software, I can write robust and failproof code to be integrated in complex Cloud Systems, with a focus on hardware and software cybersecurity.

I have worked in Home Automation, Home Security, Industrial Safety and Transport industries, which allows me to quickly get a grasp of technical stacks and business applicatives.

Skills Summary

- C/C++, Python
 - FreeRTOS, Embedded Linux: Buildroot, Yocto
 - ARM Cortex-M, Cortex-A, STM32, nRF52, Ambarella, Broadcom
 - System Architecture & Integration
 - Hardware Validation & Industrialization
 - Embedded Hardware and Software Cybersecurity
 - CI/CD, Test automation: Jenkins, GitLab CI/CD
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Professional Experience

Freelance Embedded Engineer – Kal Engineering Studio

2024 – Present

- Ported and Refactored a full **C** project to streamline product line maintenance and update costs, including porting to a **CMake** compatible toolchain.
 - Defined the hardware/software architecture for robust operation and product evolution future-proofing.
 - Created integration tests to simulate back-end operations in **node.js**, with the emulated back-end running on a **Raspberry Pi 3/Zero W**.
 - Proof of concept of a **Yocto** based autonomous robot for indoor localization algorithm validation.
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Embedded Engineer – Netatmo

2018 – 2024

- Developed firmware in **C** for a HomeKit-compatible smart door lock using BLE and NFC.
 - Participated in the product system architecture covering embedded device, mobile applications (iOS/Android) and backend services.
 - Developed an endurance test suite with **PyQt** to validate product reliability in the long term.
 - Wrote factory scripts in **Python** for product configuration and provisioning.
 - Implemented echo-cancellation algorithm on an **Ambarella/Buildroot-based** indoor camera.
 - Conducted hardware and software validation on an **Ambarella/Buildroot-based** indoor camera.
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Software Engineer – Somfy (Technical Department)

2016 – 2018

- Designed a generic **UML** software architecture for IoT actuators, with a focus on interoperability with emerging protocols.
 - **C/C++** Code generation to integrate this architecture in a solar-powered roller shutter.
 - Implemented **Jenkins-based** automated unit and integration testing using **Gcov**.
 - Setup and ran automated Static-Analysis: **LDRA**, **CodeSonar**, **CppCheck**.
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Development Engineer – Agixis (for Somfy)

2015 – 2016

- Developed critical software components in **C** on **STM32** and **EFM32** platforms.
 - Created a benchmark software for new MCU qualification.
 - Delivered technical support for the software components integration in multiple products.
 - Developed Hardware-in-the-loop integration tests.
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Embedded Systems Intern – Adetel

2015 (5 months)

- Designed and implemented **FPGA** based **HDMI/SDI/VGA** video interfaces.
 - Wrote corresponding drivers in **C** on a Microblaze softcore processor.
 - Integrated the solution into a **Zynq-7000** platform running **Embedded Linux**.
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Education

Grenoble INP – Esisar

2010 – 2015 Engineering degree in Embedded Systems (Hardware & Software)

EPFL (Exchange Program)

2014 – 2015 Micro and Nanoelectronics

Certifications and Training

Agile SCRUM

2016 – 2017 Agile SCRUM coaching sessions and application.

Embedded C++

2017 Ac6 delivered certification.

TOEIC®

2014 TOEIC® Program certification (940)

Personal Projects

- **Python Google API** and **Raspberry Pi Zero** based Alarm clock, **Raspberry Pi Zero W** with camera sensor rover.
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Hobbies

- Music, Drums, Hiking, Painting